

HAT-P-36b

R-0686

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260408 · created 2026-07-27T20:10:01+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461138.7046 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.012
Transit depth (Rp/Rs)^2	1.82 +/- 0.31 [%]
Orbital Inclination (inc)	84.6 +/- 2.04
Airmass coefficient 1 (a1)	1.227 +/- 0.013
Airmass coefficient 2 (a2)	-0.197 +/- 0.028
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	None
Optimal Aperture	3.77
Optimal Annulus	7.82
Transit Duration (day)	0.0881 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.012 vs 0.126 (deviation 0.68 σ)
Duration fit vs published	0.0881 d vs 0.0925 d (deviation 0.89 σ)
Mid-time O-C	-9.6 min
Depth SNR	10.2
Residual scatter	1.03 %

Light curve

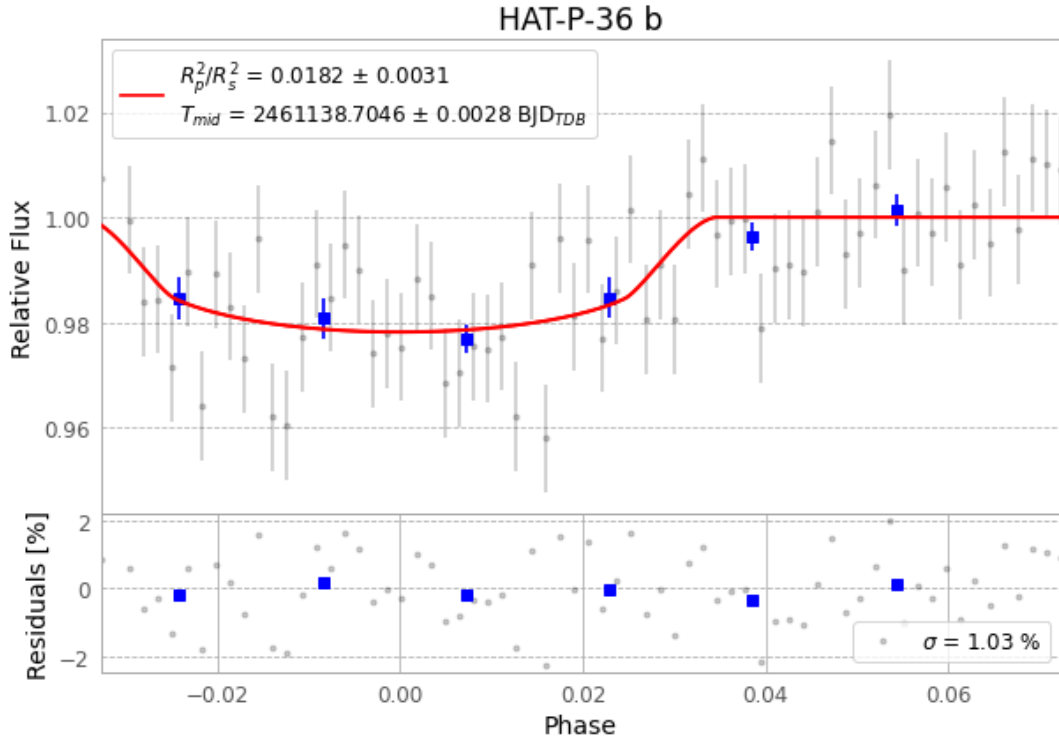


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-04-07.png (SHA-256 beef3b11c2c8a050...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [277, 243], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 be63beb0e883f834...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

68 FITS frames (45 MB), HATP-36260408034516.FITS → HATP-36260408070615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260408.log (8418e8edf64b...), FinalLightCurve_HAT-P-36 b_2026-04-07.png (beef3b11c2c8...), FinalLightCurve_HAT-P-36 b_2026-04-07.csv (be156cb25b07...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:10Z.